

CHAPTER EIGHT

LIGHT IN MEDICINE

8.1- General properties of light:

The visible light is a small band of the electromagnetic radiation (EMR) spectrum composed from discrete bundles of energy called photons.

The wavelength range of visible light $\approx 3800-7500 \text{ \AA}$ (Angstrom, where $1 \text{ \AA} = 10^{-10} \text{ m}$), and the frequency range $\approx 400-790 \text{ THz}$ (Trillion Hertz, where $T = 10^{12}$). (Fig 8.1).

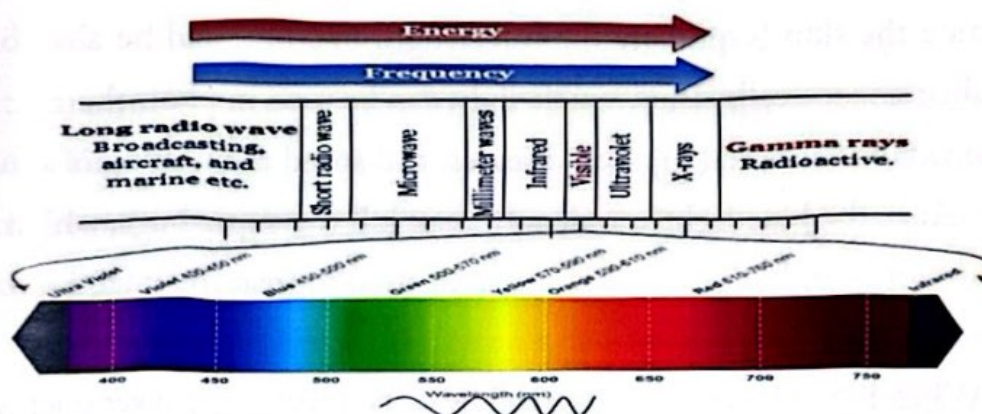


Fig 8.1: Spectrum of electromagnetic radiation.

There are some general properties of light such as:

1- The speed of light = $3 \times 10^8 \text{ m/s}$ in vacuum, and it is reduced when it is refracted from one tissue to another, i.e., it is reduced in the tissues of the body. The light beam bends (refracted) if it meets a tissue or an organ in the body at an angle other than the perpendicular angle, and its speed will be changed relative to the characteristics of the tissue or organ. Refraction of light in the eye plays very important role in vision by focusing the light on retina, the sensitive film of the eye.

2- The ratio of light speed in a vacuum to its speed in each tissue is called the index of refraction of this tissue. The body tissues and organs have different

indices of refraction. The difference of the index of refraction property permits light to be focused, and it is the reason why the optical devices (endoscopes) can be used to clearly see the objects inside of the internal body organs.

Ibn Sahl, (940-1000), Iraqi mathematician (originally Persian) invented the law of refraction, which is called later Snell's law.

3- When a light photon is absorbed, its energy generally appears as heat. This property is the basis for using a beam of visible light, or laser, to heat the body tissues by focusing the beam on the tissues, and it is utilized in heat therapy, ophthalmology, and other medical uses. [laser is very intensive monochromatic beam of light or other EMR spectrum. (Sec 9.1)].

4- Visible light of wavelengths near the infrared region of EMR can penetrate the skin (especially the wavelength 670 nm) and be absorbed by the subcutaneous cells. Thus, visible light can be used in phototherapy to act on the wounds, internal injuries, diseases, and speed the healing of long bed ulcers. Also, the blue light wavelengths can kill the super bugs, which resist the antibiotics and other drugs, and can be used to treat the infection caused by these germs.

5- When light photons are absorbed by certain materials, lower energy light photons can be emitted by these materials. This property is known as fluorescence. The melanin, the segments behind the skin color, has fluorescent component that can enhance the production of tiny amount of visible light (≈ 1000 times $<$ natural visible light) during day light (10 a.m. - 4 p.m.) especially from the face. This faint light is linked to the metabolism of the body, and the ability of detection of this faint light may play a good role in medicine in future.

6- When the light waves incident on any material, they will be reflected, scattered, or/and refracted, depending on the energy of light waves and kind of material. This gives the basis of eye vision of the materials outside the body, and also inside the body by using visual aids like mirrors, lenses, endoscopes, ophthalmoscopes, and other optical devices. (Sec 8.13).

8.2- Vision by the eye optical and neural portions:

Vision, the ability to see, is an optical process in which the eye forms images on the retina, transmits these images as signals to the brain, and the brain interprets these signals to the real figures of the images. The vision is an important sensation where nearly 70% of all the sensory receptors of the body are in the eye and about 58% of the cerebral cortex is involved in the visual aspects.

The adult normal eye is a spherical visual device with a diameter ≈ 2.5 cm, and about sixth of its surface is visible which forms the anterior of the eye.

The eye optical power (p) is the reciprocal of focal length (f) of the relaxed eye in air i.e., $p = 1/f$, where f in m. The eye power ≈ 60 D (Diopter), (where D is a unit of measuring the optical power of eye), and the focal length ≈ 16.7 mm.

The eye contains two major portions, the optical portion and neural portion.

The optical portion consists of the cornea, iris, pupil, aqueous humor, lens, and vitreous humor. (Fig 8.2a). It focuses the light on retina, a part of the neural portion, which produces electrical signals due to the interaction of light with the light receptors, the rods and cones in retina.

The normal vision in the optical portion occurs when the light of the object is focused into one focus on the retina. (Fig 8.2b).

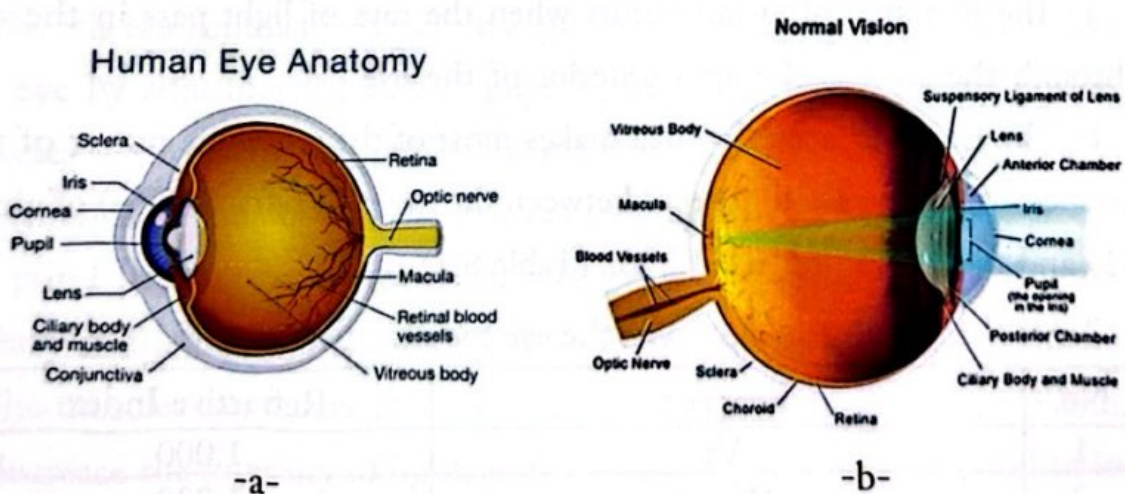


Fig 8.2: Eye vision. a- human eye anatomy. b- normal vision.

The neural portion consists of the retina, optical nerve, and visual cortex

of the brain. The electric signals produced in retina are transmitted via the optical nerve to the visual cortex of the brain, where the brain interprets these signals to real images of the objects.

1- Light focusing by the optical portion of the eye

It depends on the corneal refraction of light, control of light intensity by the iris and pupil, spread of light to the lens by the aqueous humor, lenticular refraction of light, and spread of light to retina by the vitreous humor.

I- Corneal refraction of light:

The cornea is a spherical tough transparent membrane of a diameter ≈ 11.3 mm, and a thickness ≈ 0.5 - 0.6 mm in the periphery. It covers the iris, pupil, and aqueous humor, and acts like a convex lens which can converge the light. It is constructed from six layers, the epithelium layer, Bowman's layer, corneal stroma, Dua's layer, Descemet's membrane, and corneal endothelium. (Sec 9.14.2).

The cornea has a very thin crystalized unmyelinated (transparent) nerves, and it does not have blood vessels (avascular). It gets oxygen and nutrient from two sources, the tears of outer surface and the aqueous humor at its posterior, which dissolve through the other parts of cornea.

The refraction of light through cornea can be described as follows:

a- The first step of vision occurs when the rays of light pass in the eye through the cornea, the apex anterior of the eye.

b- The front surface of cornea makes most of the refractive power of the eye because of the big difference between the index of refraction (n) of air ($n = 1$) and that of cornea ($n = 1.376$). (Table 8.1).

Table 8.1: Refractive indices of the eye focusing parts.

No.	Eye part	Refractive Index
1	Air	1.000
2	Water	1.330
3	Tear film	1.336-1.337
4	Cornea	1.376

5	Aqueous humor	1.336
6	Lens: central part	1.406
7	lens: peripheral	1.340
8	Vitreous humor	1.340

c- The optical power of cornea ≈ 40 D ($\approx$ two thirds of the eye optical power).

d- The shape, curvature, and geometrical properties of cornea, determine its functional property i.e., its refractive power.

e- The tears prevent light scattering and enhance the optical quality, in addition to their function of maintaining water content and oxygen exchange in the cornea,

f- The layers of cornea are maintained in a slightly dehydrated state which is required for the optical transparency of the cornea.

II- Iris and pupil control of light intensity:

Immediately after refracted by the cornea, the light pass through a small opening in the middle of the front eye called the pupil.

The opening of pupil is controlled by the iris, the front-colored part of the eye, which is a thin membrane located exactly after the cornea. The color of the eye (brown, blue, green ...etc.) is dependent on the color amount and distribution of the melanin pigments in the iris.

The iris is a contractile structure regulates the intensity of light that enter the eye by adjusting the size of pupil opening by two groups of smooth muscles surrounding the pupil. These muscles are the sphincter pupillae which encircle the pupil in the iris, and its function is to reduce the size of the pupil (i.e., as a constrictor), and the dilator pupillae which is running radially in the iris, and its function is to dilate the pupil (i.e., as a dilator).

The muscles can reduce the size of pupil to its minimum value, ≈ 3 mm, to decrease the intensity of light, and it can increase the size of pupil to its maximum value, ≈ 8 mm, to allow more light intensity to pass through the pupil.

Sometimes, when an individual is in a bright day light and moves suddenly to a dark place of very low illumination, he feels blind for several seconds until his eyes adapted for dark, and it is called dark adaption. In the opposite way when the individual moves from a very low illumination to a bright light, he also feels blind for several seconds until his eyes are adapted to the new bright light condition which is called light adaption.

The pupil opening is also controlled by another way called pupillary light reflex (photo pupillary reflex) where the retinal ganglion cells, in the back of the eye, primarily regulates the pupil size relative to the intensity of incident light on the retina.

III- Aqueous humor spread of light:

There is an anterior cavity in the eye called aqueous humor, between the iris and the lens, filled with a transparent liquid similar in composition to the blood plasma. The refracted light by cornea passes through the aqueous humor and incident on the lens. The refractive index of the aqueous humor ≈ 1.336 , and because the refractive index of cornea is a little higher than that of the aqueous humor, therefore the light will be slightly diffracted by the aqueous humor and spread to the eye lens.

The aqueous humor is generated and drained continuously. It is in a constant motion to maintain the intraocular pressure (IOP) of the eye to its normal value, where IOP ≈ 16 mm Hg in the adult eyes. It also supplies oxygen and nutrients to the cornea, lens, and some cells in the retina, and drains their metabolic waste through Schlemm's canal and delivers it into the blood vessels of the episclera (the outermost layer of sclera, where sclera is the white part of the eye) via aqueous veins.

IV- Lenticular refraction of light:

The eye lens is avascular (gets its nourishment from the aqueous humor), transparent, flexible, and biconvex structure enclosed in a thin elastic capsule. It is suspended between the aqueous humor and vitreous humor (gel like liquid fills the eyeball just behind the lens) by the ciliary muscles.

The lens is constructed from certain transparent proteins called crystalline proteins, which are arranged in about 20000 concentric layers, and so the lens is called crystalline lens.

The eye lens is formed in the embryo and normally stays in the eye to the end of life.

The adult eye lens has a thickness ≈ 5 mm, diameter ≈ 9 mm, and an index of refraction ≈ 1.386 in the peripheral, the less dense layers of the lens, and ≈ 1.406 in the central thick layer.

Anticlastic refraction of light can be summarized as follows:

a- When the light photons pass through the pupil, they incident on the aqueous humor, and spread on the eye lens which refracts and focuses the light on the retina.

b- The ciliary muscles can change the focal length of the lens by changing its convexity, i.e., the radius of curvature.

c- Changing the focal length of eye lens can change the accommodation power of the eye, the maximum variation in the eye power measured in diopter, which can view the objects at a large variation of distances. (Fig 8.3).

d- The refractive power of lens ≈ 20 D (about one third of the eye refractive power and half of that of cornea).

e- The optical function of crystalline lens is the maintaining of its light transparency, refraction of light, light focusing on the retina via the vitreous humor, and providing the eye accommodation power.

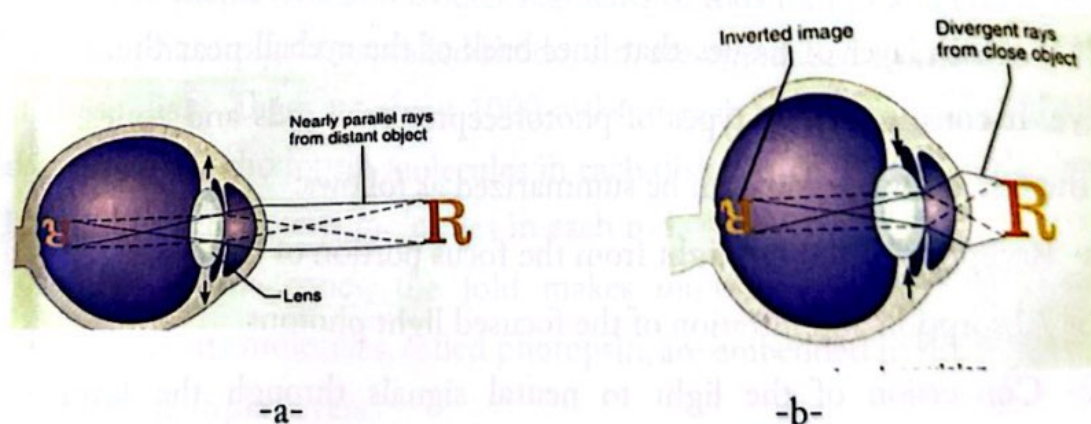


Fig 8.3: Lens focusing of light. a- lens flattened for distant vision. b- lens bulged for close vision.

V- Vitreous humor spread of light

The index of refraction of the vitreous humor ($n = 1.34$) is slightly $<$ index of refraction of the eye lens, therefore the light will be slightly diffracted by the lens and spread through the vitreous humor to the retina.

The vitreous humor filled with a transparent liquid, where the refracted light by the lens pass through and spread to focus on the retina.

2- Signals production and processing by the neural portion of the eye:

The neural portion consists of the retina, optical nerve, and visual cortex of the brain. It transmits the signals produced in retina through the optical nerve to the visual cortex of the brain.

The function of neural portion depends on the production and processing of signals in the retina.

The transmission and interpretation of these signals in the brain are concerned in the books of physiology, out of concern of this book.

The electric signals are initiated in the photoreceptors of retina, the rods and cones. The photoreceptors contain the photopigments rhodopsin and conopsin which contain the retinal, the actual part of the photoreceptors to produce the electric signals when they interact with the light photons.

I- The retina:

It is a thin layer of tissues that lines back of the eyeball near the optical nerve. It consists of two types of photoreceptors, the rods and cones.

The functions of retina can be summarized as follows:

- a- Reception of focused light from the focus portion of the eye.
- b- Absorption and filtration of the focused light photons.
- c- Conversion of the light to neural signals through the layers of photoreceptor cells of retina.
- d- Processing these signals and sending them to the optical nerve.

Essentially, the retina processes the pictures from the focused light and the brain interprets these pictures.

II- Photoreceptors (rods and cones) of retina:

The rods and cones are located on the posterior of retina, where their tips are located on the back of retina facing the choroid on the eyeball. Thus, the light enter retina must pass through all the cell layers of retina before it reaches the photoreceptors and stimulate them. (Fig 8.4).

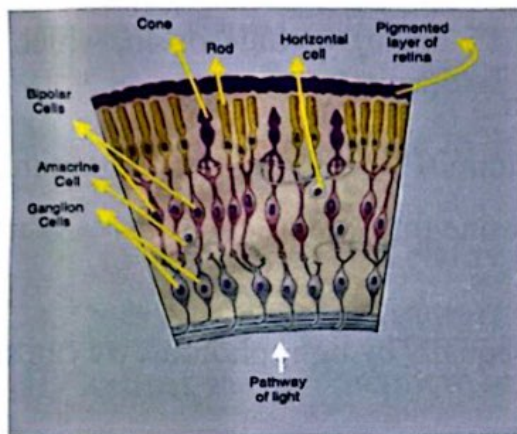


Fig 8.4: Rods and cones in the retina.

The number of rods $\approx$ 120 million, where they spread about all retina, and the number of cones $\approx$ 6 million, where most of them are concentrated in the yellow spot (fovea).

The membranes of rods and cones are layered repeatedly across their outer ends facing the choroid, farthest from light source.

The cell membrane in the outer segments of rods fold in and create disks, and the photo pigments, called rhodopsin, are embedded in the membrane of these disks. There are about 1000 disks in each typical rod, and there are about 150000 rhodopsin molecules in each disk, which give a total of about 150 million rhodopsin molecules in each rod.

While, in the cones, the fold makes multiple layers where another photopigments molecules, called photopsin, are embedded in the membrane of these multiple layers.

Because of the great availability of rods in the tissues of retina, the rhodopsin pigments are best studied than the photopsin pigments. The

rhodopsin has a peak sensitivity of rod pigment, which is very sensitive to the light, while the cone pigment has three types of cone pigments, called coneopsin, they absorb light of different wavelengths. The types of coneopsin are:

- a- The long wave sensitive opsin (L-opsin) or red opsin, which is sensitive to the red-light colors.
- b- Medium wave sensitive opsin (M- opsin) or green opsin, which is more sensitive to the green colors vision.
- c- Short wavelength sensitive opsin (S-opsin) or blue opsin, which is more sensitive to the blue colors vision.

Rhodopsin pigments are made from vitamin A, where a lack of vitamin A causes night blindness, while, missing of one or more of cone opsins causes color blindness.

When all the coneopsin are stimulated equally by light photons, we can see a white color.

In spite of their some similarities, rods and cones have some differences. (Table 8.2).

Table 8.2: Comparison between rods and cones.

No.	Rods	Cones
1	Have rod shaped structures	Have cone shaped structures.
2	About 120 million rods spread all about the boundary (peripheral of retina)	About 6 million concentrated at the center of retina especially in the fovea (yellow spot) of the retina.
3	Very sensitive to the light (one photon can make the stimulation) and thus enable vision in low intensity of light, slow response, and long integration time.	Less sensitive to the light (several photons can make stimulation) thus enable vision in only high intensity of light, fast response, and short integration time.

4	Responsible for dim light and night vision and known as scotopic vision.	Responsible for bright light and day light vision and known as photopic vision.
5	More sensitive to the light of long wavelengths (about 570 - 750 nm)	More sensitive to the light of short wavelengths (about 420, 534, and 563 nm).
6	Not responsible for the vision of details, colors and rapid changes in the images, and the objects are visible but in dark colors.	Responsible for the vision of fine details, and colors and rapid changes in the image, and the objects are visible in colors.
7	Have a low spatial clarity of vision (visual acuity)	Have a high visual acuity.

III- Retinal, the chromophore of rhodopsin and conopsin:

Retinal (was called retina or retinal aldehyde) is one form of vitamin A, with different number from one species to another. Vitamin A is stored by the liver and released to the blood when it is required by the photoreceptors to make their visual pigments.

The rhodopsin and conopsin pigments contain the retinal which is bound to the opsin, a polypeptide chain, where retinal is the chromophore of these pigments. The chromophore is a chemical group that absorbs light at a certain frequency, and so it gives a color to the molecule. The chromophore is the actual light sensitive part of the photopigment.

The chromophore is the same in the four types of pigments (one type in the rods and three types in the cones), and the differences between these four types photopigments are relative to the type of opsin binding with the retinal, and the way of this binding.

Each type of these photopigments filters the light differently and absorbs the light most effectively at certain wavelengths of the visible light spectrum.

For example, when one type of photopigment absorbs wavelengths in the range of blue light more effectively, another type of pigments bestly absorbs the red light.

The retinal (chromophore), the actual sensitive part of the photoreceptor to the light, is perpendicular to the rays of incoming light in order to catch the maximum light photons incident on it.

The retinal will detach from opsin when it absorbs light photons. This will initiate a whole chain of chemical and electrical reactions in both rods and cones. The electrical signals are produced in this reaction in a steady rate of 20-30 signals/s, whether it is a dark or well illuminated field. These signals are transmitted to the visual cortex of the brain via the optical nerve and interpreted to the real original images.

8.3- Eye defects:

Eye defects are malfunctions or problems in a certain part or parts of the eye that affect the vision.

In a normal eye, the light from an object passes through the cornea which refracts the light to the eye lens. The light will pass through the lens, refracted again, and focused on the retina. The image is formed on the retina, which is small, inverted (upside down), and real (the actual rays travel to form the image). The image is transmitted to the brain and interpreted to its real size and figure. (Fig 8.5).

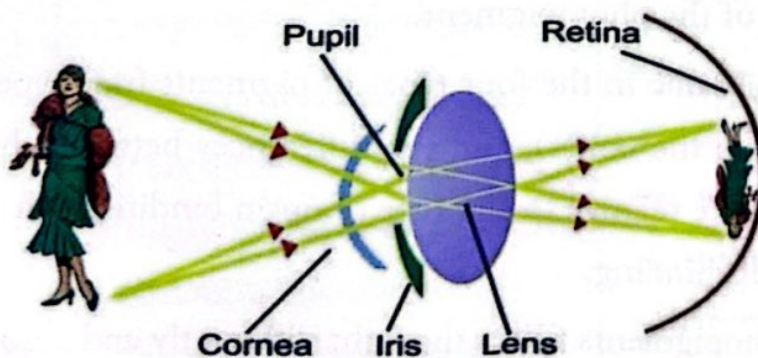


Fig 8.5: Image formation in the normal eye.

The eye structure is designed cleverly in a normal shape which is very important to focus the object rays on one focus on the retina.

The normal eye can see distinctly between 25 cm (near distinct point) and infinity (far distinct point).

For some people, the eye shape is abnormal, and the object rays are focused far from the retina (in front or behind retina), or on the retina but in more than one focus, which are called eye defects.

The defects are not due to a pathological disease, but they are related to the structure of the optical system of the eye such as the cornea, eye lens, eyeball...etc, which can affect the focal length of the eye, visual acuity, and accommodation power of the lens.

There are several types of eye defects, but the most common of them are six defects which are the myopia (short sight, short sightedness, or near sightedness), hypermetropia (hyperopia, long sight, long sightedness, or far sightedness), presbyopia (old sight), astigmatism, keratoconus, and cataracts defects.

8.4- Myopia (near sightedness) defect and its correction:

It is an eye defect in which the person can see the near objects clearly while the far objects appear blurred, i.e., the myopic eye, either has a near distinct point < 25 cm, a far distinct point $< \infty$, or it has both.

The reasons of myopia defect are that the cornea has a curvature more than normal, eyeball is too long, or eye lens is more rounded, and in some cases, it is a combination of two or more of these factors.

Myopia can often be regarded as the first sign of type II diabetes.

1- Myopic eye vision:

It can be described as follows:

I- The eye refractive power is stronger (> 60 D), and the focal length of the eye is smaller (< 16.7 m) than that of the normal eye.

II- Because of the excess refractive power, cornea cannot focus the light

rays on the retina but at a point in front of the retina. (Fig. 8.6a).

III- Myopic person has a difficulty to clearly see the distant objects and reading road signs. Also, he feels fatigue, eye strain, squinting, and headache when he is playing sport or driving. But he can see well for reading at short distances such as the computer use.

2- Correction of myopic vision:

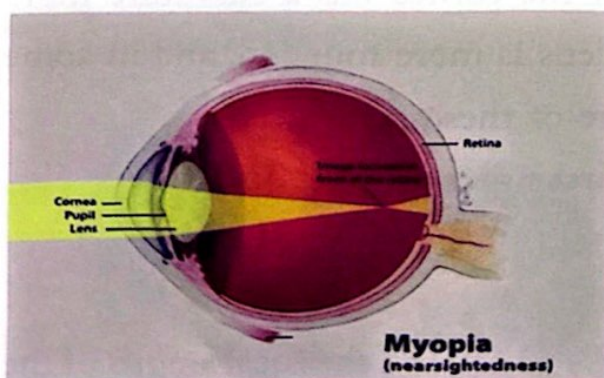
The correction of myopic vision can be summarized as follows:

I- Myopia defect can be examined using a device called retinoscope. (Sec 8.12.4).

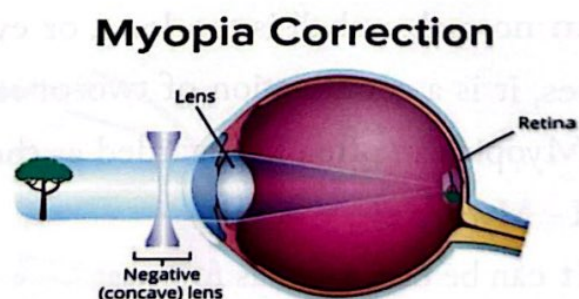
II- The prescription of myopic patient is a spherical concave eye glasses (spectacles) with a minus powered (-ve sign precedes lens power number). The higher negative number indicates that the patient is more near sighted. The concave lens is a part of a spherical shape which is thinner at the center and thicker at the ends. (Fig 8.6b).

The first one who invented the glasses to help the weakened sighted people is Hasan Ibn al- Haytham, (965-1040), Iraqi physicist, mathematician, and astronomer.

III- When a myopic person is wearing his glasses and he still feels these symptoms, therefore he requires a stronger prescription. This prescription can be determined by using a device called phoropter (refractor). (Sec 8.12.5).



-a-



-b-

Fig 8.6: Myopia eye defect and its correction. a- myopia defect. b- myopia

spectacles correction.

A corrective concave lens of an appropriate power should be selected to move, as more as possible, the near point to 25 cm and the far point to infinity.

IV- The required power P of the corrective diverging lens can be calculated using the thin lens formula:

$$(P = \frac{1}{f} = \frac{1}{u} + \frac{1}{v})$$

Where f is the focal length of the concave lens, u is the object distance, and v is the image distance from the eye (≈ 0.02 m). The unit of P is D and that of u and v is m.

V- Also, contact lenses (sec 8.10) and laser surgeries such as lasik (sec 9.15), lasek (sec 9.16), or PRK (sec 9.17), can be used to treat the myopia.

VI- It is rarely happened that a rapid elongation of the eyeball occurred at childhood which leads to a sever progress of myopia and loss of vision. This case is called degenerative or malignant myopia, which increases the risk for bleeding in the eye, cataracts, retinal detachment, and blindness. A combination of drug and laser procedure called photodynamic therapy is used for the treatment of degenerative myopia or slowing down its growth. (Sec 9.12).

8.5- Hypermetropia (far sightedness) defect and its correction:

It is an eye defect in which the person can see the far objects clearly while the near objects appear blurred, i.e., the hypermetropia eye has a near distinct point of vision > 25 cm and far point $= \infty$.

1- Hypermetropia eye vision:

It can be described as follows:

I- The reasons of hypermetropia defect are that; the curvature of cornea $<$ curvature of normal eye, the eyeball is too short, or the eye lens is flatter.

II- The lens has a low converging power because of the weak action of the ciliary muscles which control the lens contraction, where the refractive

power of the eye is weak, and the focal length of the eye > 16.7 m.

III- Because of the weakness in the refractive power of the eye, the light rays cannot be focused on the retina. The geometrical extension of the light rays (imaginary lines) makes the focus behind of the retina. (Fig. 8.7a).

IV- The signs and symptoms of hypermetropia are the eye strain (which is the common symptom), blurry vision, headaches, squinting, and the patient feels fatigue when he performs a work at a close distance.

V- The risk factors of hypermetropia include the family history of this defect, tumors around the eye, diabetes, and certain medications.

2- Correction of hypermetropia:

It can be summarized as follows:

I- The hypermetropia can be examined using a retinoscope and can be corrected by using spherical convex (converging) glasses (+ve power lenses).

II- The corrective lenses should have appropriate powers or focal lengths that can supply the required additional focusing power to bring the image of the object on the retina. (Fig 8.7b).

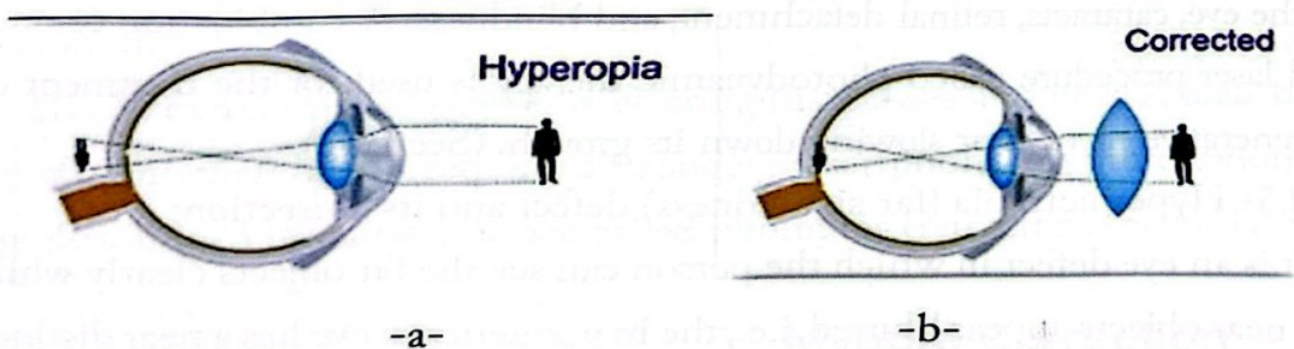


Fig 8.7: Hypermetropia eye defect and its correction. a- hypermetropia defect. b- hypermetropia spectacles correction.

III- The optician can examine the eyes and write the required prescription for the defected eyes, and the lens power is calculated in the same manner as that of the myopia except it is +ve. The prescription of hypermetropia defect can be determined by using a phoropter.

IV- Nowadays, the required eyeglasses (lenses) for correction of the

farsightedness defect should be selected as follows:

- a- An aspherical lens (not a part of spherical or cylindrical lenses).
- b- Coated with anti-reflecting coating to eliminate the reflection of light from the lenses, and the other people can see the eyes of the wearer.
- c- Made with flatter curves than that of the normal (conventional) lenses. They are thinner, lighter, slimmer, more attractive profile, and can eliminate the spherical aberration and reduce other aberrations. While the conventional (convex) lenses used to correct the hypermetropia are thicker in the center and thinner at the edges.
- d- Even for stronger prescriptions, the aspheric plus lenses are much thinner at the center and there is no bulging of the lens from the center which gives the wearer a slimmer and more flattering profile. While the center of conventional convex lens is thicker and bulges forward from the frame of the glasses.
(The aspherical lenses are also used in myopia to make thinner lenses, but they are not important as that of the long sightedness defect).
- e- Contact lenses, lasik, lasek, or PRK, can be used for correction of the hypermetropia.
- f- In children, hypermetropia is usually genetic (family history) and many children are born with hypermetropia, but it can correct itself with time because the eyeball lengthen and grow with time. If the hypermetropia eye is not corrected in the young children, it will stay weak and never see as well as the normal eye.

8.6- Presbyopia (old sightedness) defect and its correction:

It is a far sightedness defect associated with aging in which the near objects appear blurred. The eye lens loses progressively its ability to focus clearly on the near objects, either by losing of elasticity of the ciliary muscles that control the eye lens movements, or by increasing the stiffness and hardening of the eye lens. This will result in focusing of the light out of retina, imaginary behind the retina (in the same way as in the hypermetropia).

It is common that a patient with presbyopia can have another type of refractive defect at the same time, such as myopia, hypermetropia, or astigmatism.

Presbyopia diagnosis can be done by eye exam using a retinoscope, and the prescription of glasses to correct the presbyopia defect can be determined by using a phoropter.

Presbyopia can be treated typically by using bifocal, progressive additional lenses, monovision lens, or contact lens.

1- Bifocal (two points of focus) lens, in which there are two main parts of the lens with a visible line between them. The main upper part of the bifocal lens is used for distance vision, while the lower part of the lens is used for close work.

2- Progressive additional lenses (PALS), which are similar to the bifocal lenses, but they offer more gradual visual transition, without the visible line between the two main parts of the bifocal lens.

3- Mono vision lenses, in which one eye wears a lens that aids in near vision while the other eye has a lens that aids in far vision.

4- Contact lenses, where soft or gas permeable lens can be used. (Sec 8.10).

8.7- Astigmatism defect and its correction:

The astigmatism defect is an eye refractive defect that the cornea or lens do not focus light in one focus on the retina, but it falls in multiple foci, which results in a blurred vision at all distances from the eye. (Fig 8.8a).

The first time that astigmatism reported was at the beginning of 19th century by the British polymath and physician, Thomas Young, (1773-1829).

In order to understand the astigmatism defect and its correction, this subject will focus on the characteristics of astigmatism defect, eye meridians (imaginary lines), types of astigmatism, astigmatism correction, and astigmatism prescription.

1- Characteristics of astigmatism defect:

The astigmatism is usually described by:

I- Irregular curvature of cornea (corneal astigmatism), in which the cornea does not have a spherical shape, but it has aspherical (not regular spherical) shape, or its shape like the shape of a football (prolate spheroid).

II- Change of the lens shape (lenticular astigmatism), in which the lens shape has different convexity. In astigmatic eye, the focal lengths of cornea and eye lens for an object are oriented in some directions, which are not the same as for another object located in a perpendicular direction.

III- The symptoms of astigmatism include:

- a- Eye strain (asthenopia) blurred vision.
- b- Fuzzy vision distorted at all distances.
- c- Difficulty of seeing at night.
- d- Squinting, eye irritation, and pain in and around the eye.
- e- Headache feeling especially after a long time of visual task like reading, watching TV, or working on a computer.

2- Eye meridians:

They are two virtual lines corresponding to the greatest circumference of the eye through the interior and posterior poles of the eyeball.

When front of the eye is imagined as a phase of the clock, the first eye meridian, the horizontal meridian, can be considered as the line between 9 and 3, and the second meridian, the perpendicular meridian to the horizontal meridian, can be considered as the line between 12 and 6.

The horizontal and vertical meridians of the corneal astigmatic eye, football shaped cornea, are different in curvature where one meridian is significantly more curved than the meridian perpendicular to it. This results in a well focusing of objects on one meridian and not well focusing on the other meridian, and the steeper and flattered meridians are called the principal meridians.

A patient with astigmatism cannot simultaneously focus on both horizontal and vertical meridians, and he can see one meridian clearer than the other.

The meridians in astig. are similar to that of the imaginary longitudinal lines of the globe.

3- Types of astigmatism:

There are three types of astigmatism which depend on the defected principal meridians, and other types depend on whether the principal meridians are perpendicular or not. These types are :

I- Myopic astigmatism, in which one or both principal meridians of the eye are near sighted.

II- Hyperopic astigmatism, in which one or more principal meridians are far sighted,

III- Mixed astigmatism, in which one principal meridian is a near sighted and the other is a far sighted.

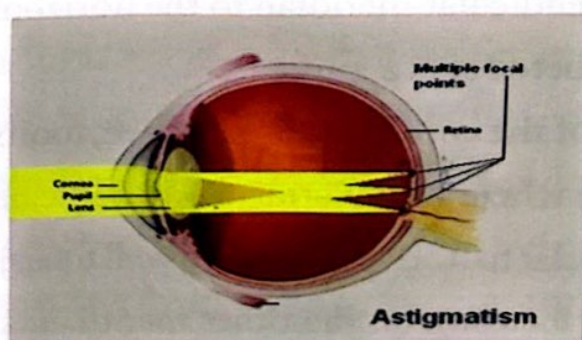
IV- Regular astigmatism, in which the principal meridians are perpendicular. Most kinds of astigmatism are regular corneal astigmatism, where the eye surface is shaped like the shape of a football.

V- Irregular astigmatism, when the principal meridians are not perpendicular, which results normally from an injury that causes a gradual thinning of the cornea.

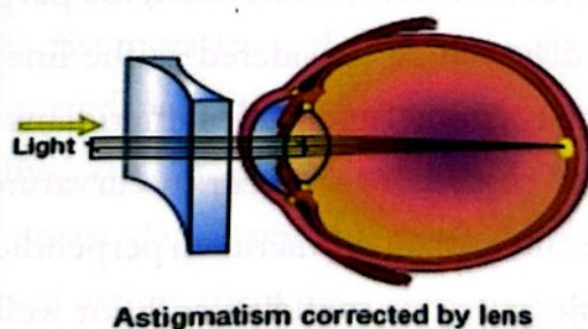
4- Astigmatism correction:

Correction of astigmatism can be summarized as follows:

I- The conventional treatment or correction of astigmatism can be done by the eyeglasses using cylindrical lenses in addition to spherical lenses. (Fig 8.8b).



-a-



-b-

Fig 8.8: Astigmatism defect and its correction. a- astigmatism defect. b- astigmatism spectacles correction.

II- The spherical lens is used to correct the near sighted in the myopic astigmatism (-ve lens) or the far sighted in the hyperopic astigmatism (+ve lens).

III- The cylindrical lens power is used to correct the difference between the powers of the two principal meridians of the eye.

IV- The axis of astigmatism in the eye glass is the location of the flatter principal meridian of the eye using 180° rotary scale, and the axis of astigmatism = 90° .

V- When the vertical meridian has a higher near sighted than the horizontal one, therefore, the thickness of the corrective glasses is more at the edge of the vertical meridian than at the edge of horizontal meridian.

VI- Other ways are used for correction of astigmatism such as contact lenses, which can provide a wider field of vision, and laser refractive surgery (lasik, lasek, and PRK) that can change the corneal shape.

5- Astigmatism prescription:

Astigmatism prescription can be written as follows:

I- The optician writes the prescription of astigmatism by using a phoropter to determine the powers of the spherical and cylindrical lenses in addition to the axes of astigmatism.

II- The calculation of power of the spherical lens is based on the thin lens formula, but that of the cylindrical lens based on more complicated formula.

III- When a prescription, for example, of myopic astigmatic eye of a patient was written as - 4.5 -2 x 90° , therefore:

The first number (-4.5) represents the spherical lens power, the second number (-2) represents the power of cylindrical lens, and the third number (90°) is the axis of astigmatism.

Example 8.1:

If the prescription of an astigmatism patient is $-5.00 - 1.00 \times 180^\circ$, calculate the amount of astigmatism.

Solution:

From the prescription

The patient needs a lens power of -5 D to correct his vision in one meridian.

In the other meridian, the patient requires a lens power = $-5 - 1 = -6$ D.

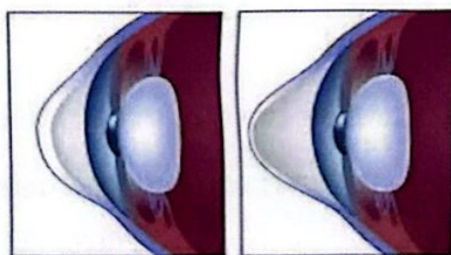
Thus, the amount of astigmatism equals the difference between the powers of the two meridians i.e.,

$$\text{Amount of astigmatism} = -6 - (-5) = -1 \text{ D}$$

8.8- Keratoconus defect and its correction:

Keratoconus (KC, KCN, or KTCN) defect, also called steep cornea, is a degenerative disorder of the eye cornea occurs when the cornea thins, its shape is gradually changed from the spherical shape to a conical shape, and bulges outward. It causes blurred vision, and glare.

The conical shape of cornea is a result of the collagen deficiency in the protein tiny fibers that hold the cornea. They are weakened and cannot hold the spherical shape of cornea which becomes progressively more cone shaped. (Fig 8.9).



Normal

Keratoconus

-a-



-b-

Fig 8.9: Keratoconus eye defect. a- normal cornea and cornea with keratoconus defect. b- outlook of a keratoconus defected eye.

These changes in cornea can cause blurred vision, streaking of light,

sensitivity to the light, glare, and halos (seeing bright circles or rings around the light source).

Keratoconus can be diagnosed by several methods such as:

1- Slit lamp test, (sec 8.12.2), to inspect the central or inferior corneal thinning.

2- Corneal topography, (sec 8.12.8), for detection and following the early cases of keratoconus.

3- Algorithms, using the corneal topography to detect the forme fruste (incomplete form or partial symptom of a disease) keratoconus i.e., the cornea that has no abnormal findings when both the slit lamp and corneal topography are used. The algorithm here is a finite list of instructions which are given in order to solve the curvature of cornea (as a complex problem) by computer programming where the data are obtained by using corneal topography.

4- Ultrasound pachymetry (a test to measure the thickness of cornea) where ultrasound waves of frequency = 10-20 MHz are traditionally used to measure the thinnest zone of cornea.

Keratoconus can be corrected by spectacles (for mild cases), contact lenses [rigid gas permeable (GP) and hybrid lenses], and transplantation of cornea (for severe cases) which requires contact lenses afterward.

Lasik is dangerous for people with keratoconus because it can weaken the cornea more and make the vision worse.

8.9- Cataract defect and its correction:

Cataract defect is the loss of transparency and the ability of accommodation of the eye lens which leads to a decrease in the vision or blindness. It occurs in either one or both eyes.

The symptoms of eye cataract include blurred clouded or double vision, difficulty of vision at night, yellowing or fading of colors, seeing halos around the light, and sensitivity to light, and glare. But the most common reason of cataract is the aging.

The surface cells of lens are the only cells that retain the capacity to divide, and when new cells are formed, the older cells go deeper in the lens. Thus, the central part of the lens becomes stiffer, and its color changes from yellow to black gradually with age. This causes opacity of eye lens, which is responsible for the eye cataract, the most common eye defect. Also, some of the protein in the lens may clump together and start to cloud a small area on the lens. This cloud may grow larger and larger on the lens making it harder to see.

There are other reasons of cataract such as diabetes, past eye surgery, long term intake of steroid medication, smoking, alcohol, trauma, hypertension, obesity, high myopia, injuries, and radiation.

Cataract cannot be corrected by glasses or corrective surgery, but it can be corrected by very safe and effective surgery, which is one of the most common, safe, and effective surgery in the world.

In this surgery the natural eye lens is removed and replaced by a man-made (artificial) lens, which is called intraocular lens (IOL), to restore the clear vision.

Even the effective vision can be restored with the aid of the implanted artificial lens, but the ability of the artificial lens to accommodate is getting lost.

Laser assisted cataract surgery is also developed which offers more precision, less trauma and complications, and faster recovery. (Sec 9.18).

8.10- Contact lenses:

The contact lens (CL) is a thin plastic lens, considered as a biomaterial medical device, placed directly on the eye surface to correct vision, or for cosmetic reasons. It corrects vision like the eye glass, and it is safe when used with care. It is more ideal than the normal glasses for sports and other outdoor activities because it does not collect moisture or perspiration.

The natural irregular shape of cornea can be replaced by the contact lenses in the patients who have sight defects or distorted corneal shapes, as those with keratoconus, to provide a good level of vision.

The idea of contact lens was first introduced in 1508 by the Italian polymath Leonardo Da Vinci, (1452-1519), when he described a method directly changes the cornea's power by wearing a hemisphere water-filled glass over the eye. But Adolf Fick, (1852 - 1937), German ophthalmologist, successfully constructed and fitted the first contact lens in 1888.

There are several types of contact lenses used for correction of the eye defects, where the most important of them are the toric, gas permeable, bitoric gas permeable, and hybrid contact lenses.

1- Toric contact lens:

It is a soft lens used to correct astigmatism and gives balanced and clear vision to the patient. The geometrical shape of the toric lens is that it has the form of torus or parts of torus. It has two different surfaces instead of one surface of the lens, one phase looks like a cap from a torus (a side slice of a donut), and the other surface is usually spherical. (Fig 8.10).

The focal length and optical power in the toric lens are in two meridian orientations perpendicular to each other.

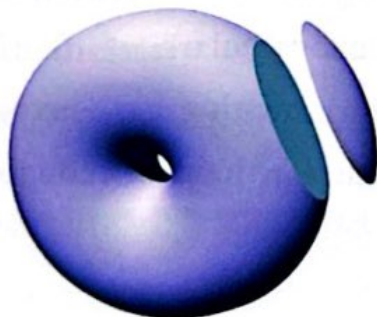


Fig 8.10: Toric lens, a cap surface (the upper right part in the image).

It behaves like a combination of two lenses, spherical and cylindrical lenses, for this it is suitable to correct the astigmatism defect. The toric lens has different optical powers and focal lengths in the different meridians. It has a design feature that enables the lens to rotate to the proper orientation on the cornea to align the power meridians of the lens with the appropriate meridians of the defected eye for a clear vision. For best visual acuity, (see 8.11), the toric lens must remain in one position.

Toric lenses are more expensive than the regular spherical contact lenses because its design, fitting in the eye, and its complex exams. Toric lenses rotate while blinking which make the vision temporarily blurred.

In some cases, the proteins secreted in the tears of the eye cause inflammatory reaction which causes lid glands to secrete certain substances. These substances can create a protein film on the soft contact lenses making them uncomfortable and cause vision problems. Thus, disposable soft contact lenses are normally used for single day of wearing, because there is not enough time for proteins to accumulate on the disposable lenses.

2- Rigid gas permeable (RGP or GP) contact lens:

The cornea must be transparent in order to function properly. It is avascular (no blood supply) and absorbs oxygen from the tear film. Thus, high oxygen permeable contact lenses should be used, so that, the cells of cornea absorb the required amount of oxygen.

Hard contact lenses reduce the oxygen absorption of corneal cells. Sometimes, this will result in several uncomfortable conditions like swollen cornea, red eyes, blurred vision, risk of developing an eye infection, and growth of blood vessels in the cornea (corneal neovascularization).

Some modern contact lenses are made of a material with greater oxygen permeability, called rigid gas permeable lenses, that can help in reduction of the risk of corneal neovascularization. These lenses are made either from a durable plastic that transmits oxygen, or soft silicon hydrogel which are made from a gel like the plastic material. (Fig 8.11a).

There are two fundamental advantages of GP lenses which are:

- I- The tears do not adhere with this kind of lenses.
- II- They can form a tear film of the eye into a spherical shape, which will correct the astigmatism and result in a very good vision.

GP lenses can mostly correct astigmatism without a toric design because it can retain its spherical shape on the eye and replaces the irregular shapes of cornea.

Also, there are some toric GP contact lenses used only for the severe or unusual types of astigmatism which are not common.

Even though, GP lenses are an ideal solution for astigmatism, the patient requires an adaption period in wearing GP lenses before he feels that these lenses are comfortable as the soft lenses.

3- Bitoric GP contact lens:

A bitoric GP lens has a toric shape on both sides, and it has two different curves, one on the front surface and the other on the back surface. (Fig 8.11b).

When the corneal cylinder power ≥ 2.5 D, another kind of GP contact lenses are designed, called bitoric GP lenses, to treat this severe case. The curves of bitoric GP lens can provide an alignment fitting relationship in both meridians and gives a good vision.

These lenses are prescribed for high amounts of corneal astigmatism i.e., high variation of corneal curvature between the flat and steep meridians which are perpendicular to each other.

4- Hybrid contact lens:

The hybrid contact lens is a large diameter lens that consists of a GP lens in its central zone surrounded by a soft contact lens, which gives a great correction for corneal astigmatism. (Fig 8.11c).

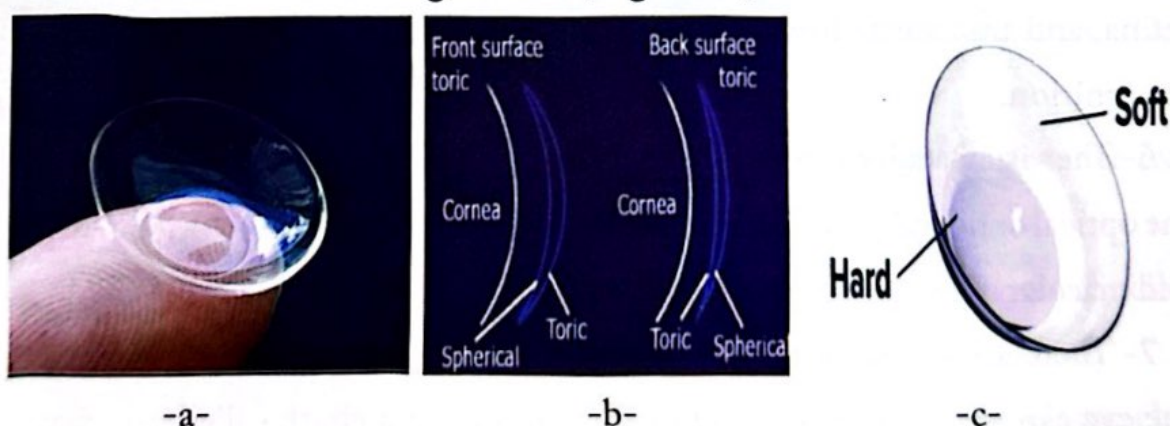


Fig 8.11: Rigid gas permeable lenses. a- GP lens. b- bitoric GP lens. c- hybrid GP lens.

The hybrid contact lenses can provide a good vision by the GP lens which

is able to mask most of the corneal astigmatism combined with the wearing comfort of the soft contact lens.

Several types of hybrid lenses, including multifocal design, are used to correct myopia, presbyopia, astigmatism.

A patient prescribed a hybrid contact lens will find the vision is good, wearing of lenses is comfortable, and the price is pleasing. But he requires a training for proper insertion, removal, and caring of his hybrid lenses.

8.11- Visual Acuity:

It is the clarity of vision, or the ability of eye to perceive small details.

Visual acuity depends on several factors such as:

- 1- Ability of the optical portion of the eye to focus the light sharply on retina.
- 2- Sensitivity of rods and cones to the focused light to produce signals.
- 3- Ability of optical nerves to transmit the signals to the brain.
- 4- Capability of the brain to interpret these signals.
- 5- The normal (maximum) visual acuity is obtained when the optical and neural portions function properly. This means that the outside world is projected as a sharp image on the retina, translated as neural impulses by retina, and transmitted to the brain via the optical nerve for analyzing and recognition.
- 6- The visual acuity is reduced when there is a disorder in a part or parts of the optical or neural portions like a sight defect, retinal detachment, glaucoma, and macular degeneration. Also, diabetes can affect the visual acuity.
- 7- There are several ways to test the visual acuity, which means how well the eye can see the details of letters or symbols in a chart, called eye chart, from certain distances.

8.12- Eye Vision test devices:

There are several important devices used for vision and eye test such as the

eye charts, slit lamp, tonometer, retinoscope, phoropter, ophthalmoscopes, keratometry, and corneal topographer.

1- Vision Charts:

They are portable plastic charts used for measurement of visual acuity and vision sharpness which employ certain types of letters or symbols. They are normally used by the physicians, nurses, or optometrists to screen the patient for vision impairments (range from very low vision to the inability to see colors). Also, the ophthalmologists use the eye chart for monitoring the visual acuity for their patients in therapies of surgery or medication.

The most widely used types of eye charts are the Snellen chart, Landolt C chart, E chart, and logMAR chart where certain optotypes (standardized symbols) are utilized in these charts.

I- Snellen's chart:

Snellen's chart is a vision test used to quickly measure the monocular and binocular visual acuity of the eye, by determining the level of visual details of certain capital letters that the patient can discriminate.

According to the British Standard Institution (BSI) in 1968, only nine capital letters C, D, E, F, L, O, P, T and Z are used in a variety of randomized patterns. In 2003, these letters were replaced by 12 letters, which are C, D, E, F, H, K, N, P, R, U, V and Z.

Snellen chart is normally placed on a wall with 6 m (20 ft) distance from the patient.

It was developed in 1862 by the Dutch ophthalmologist, Herman Snellen, (1834 – 1908), and it is named after him.

There are different types of Snellen chart, but basically, it is a plastic chart with 11 rows (lines) of capital letters. The above row consists only one letter, usually a big E, and the other rows contain other letters which are getting progressively smaller in lower rows. (Fig 8.12).

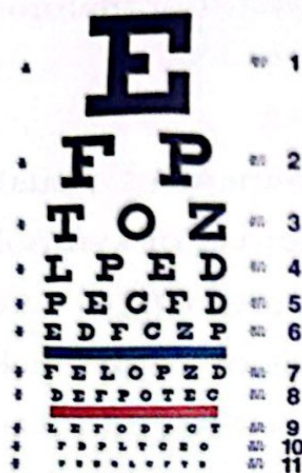


Fig 8.12: Snellen's chart.

Some Snellen's charts are designed on a different number of rows and the visual acuity is written at each line.

a- Procedure of Snellen's chart:

Mostly, the eye doctor rooms are small with a length < 6 m, therefore, the Snellen's chart is hanged after the patient, and mirrors are used to make the chart appears in front of the patient at a stimulated distance of 6 m.

To start the test of visual acuity, the patient should cover one of his eyes (e.g., the right eye) by his hand or any other cover, to see the top rows of the chart, and he will say if he can see the kind of letters or symbols in the rows. Then the test is repeated in the eye and the same manner to see the other down rows, and the last row that his eye can see will be his visual acuity of this left eye.

Then, he should close the left eye, and the same test will be done for his right eye. Of course, the visual acuity is not always the same for both the left and right eyes.

b- Measurement of visual acuity by Snellen's chart:

The letters in Snellen chart have a particular simple geometry where:
 Thickness of lines = thickness of white spaces between lines. Also:
 Height and width of the letter = $5 \times$ thickness of the line.

The visual acuity (VA) can be calculated from the Snellen's chart (and Landolt, and E charts) using the formula:

$$VA = \frac{d}{D}$$

Where d = distance from the chart to the patient (6 m or 20 ft), and D = distance at which the person with normal eye can see the letter or symbol clearly.

The visual acuity of a normal eye is represented by a fraction 6/6 (or 20/20) which means that the patient can normally see the letters (or symbols) of the chart at a distance = 6 m from the chart, as the normal eyes of other people can see.

The visual acuity of the newborn infant $\approx 6/13$ which develops to 6/6 after the age of 6 months for most of the children.

The 8 upper rows of Snellen's chart represent the visual acuity of normal eye (6/6 or 20/20), and the 3 lower rows (9,10,11) represent a visual acuity $> 6/6$ (20/20), i.e., $\frac{6}{5}, \frac{6}{4}$ and $\frac{6}{3}$ ($\frac{20}{15}, \frac{20}{12}, \frac{20}{10}$) which is called supervision.

II- Landolt C chart (Landolt ring or Landolt broken ring):

Landolt C chart, also called Japanese vision test, is a vision test using a single optotype placed in different positions, right, left, up, down, and 45° positions in between.

It consists of ring symbols with small gaps (like the letter C) arranged in rows, and these rings get smaller gradually in the lower rows. It consists of the same number of rows as in Snellen's chart, but all the letters replaced by the letter C. (Fig 8.13).

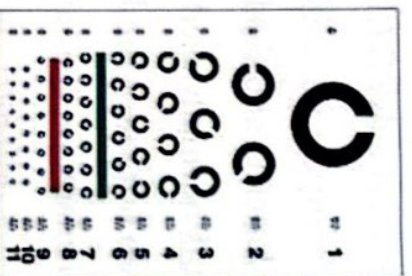


Fig 8.13: Landolt C chart.

Landolt C chart is more precise and reliable than Snellen's chart, so it is used for the driving license test. It is the standard eye chart for measuring the eye acuity in most of European countries. The procedure of test is done in the same way of that of Snellen's chart, and the patient should decide the direction of the gap. Also, the thickness of lines = thickness of the gap in the letter C, and the visual acuity can be calculated in the same way as that of Snellen's chart.

Landolt C chart was developed by the Swiss ophthalmologist Edmond Landolt, (1846 – 1926).

III- E chart (tumbling E chart):

It is also developed by Herman Snellen where the letter E is used as optotype in all the rows of the chart. The fingers of E letters are positioned facing different directions, left, right, up and down. (Fig. 8.14).

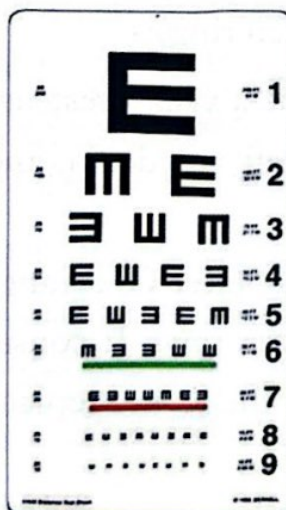


Fig 8.14: E chart.

The patient decides where the facing of fingers of letter E, and the last row that the patient can see is the visual acuity. This test is normally used for the children, illiterate patients, and other patients who do not know the English alphabetical. The visual acuity of E chart can be calculated in the same way as that of Snellen's chart.

IV- LogMAR chart:

It is an acronym for the "Logarithm of the Minimum Angle of Resolution" (where the logarithm is of the base 10). It is also called Bailey-Lovie chart of Early Treatment Diabetic Retinopathy Study (ETDRS) chart.

LogMAR chart is a visual test consisting of 15 lines, where each line has 5 capital letters, and each letter has a score value of 0.02 log units. The size of letters and the minimum angle of resolution (the ability of retina to distinguish between two adjacent letters) in each line decreased gradually from the upper to the lower line. (Fig 8.15).



Fig 8.15: LogMAR chart.

It is more accurate than the other procedures used for the measurement of the visual acuity, but it takes more time to do this test, and the calculation of logMAR value is more complex. For this, it is used to measure the VA for very important purposes such as the driving license, visual medical research like diabetes retinal detachment and other visual neural malfunctions, and routinely in children VA measurement.

a- Measurement of visual activity by logMAR chart:

The VA of a patient can be measured by the logMAR chart using the following equation:

$$\text{LogMAR VA} = 0.1 \text{ NL} + 0.02 \text{ nl} \quad \dots(8.1a)$$

Where NL = number of lines completely read by the patient, and nl = number of additional letters completely read by the patient.

LogMAR chart is scored on a basis called (by letter) where each letter can be read by the patient contributes to the final score of VA. (Table 8.3a).

For children VA test, logMAR chart is used, called Thomas pediatric screening test chart, in which four letters are displayed in each line, each letter has a score of 0.025, and gives a line score = $4 \times 0.025 = 0.1$.

LogMAR chart was developed in 1976 at the National Vision Research Institution of Australia.

b- Comparison of LogMAR with Snellen's chart:

LogMAR can be compared with Snellen's chart using the formula:

$$\text{logMAR} = -\log(\text{decimal acuity}) \quad \dots(8.1b)$$

Where the decimal acuity is that of Snellen's chart, i.e., when the acuity in Snellen's chart = 6/12, this means that the acuity decimal = 0.5, and when the acuity = 6/30, the acuity decimal = 0.2 and so on.

Example 8.2:

Calculate the corresponding logMAR values when patients have Snellen scores of 6/6, 6/19, and 6/60.

Solution:

For Snellen score 6/6, the decimal acuity = 1, therefore:

$$\text{logMAR} = -\log 1. \text{ Thus: } \text{logMAR} = 0.$$

This means that 6/6 (20/20) in Snellen chart corresponds to 0 on logMAR chart.

When Snellen chart = 6/19, the decimal acuity = $6/19 = 0.32$. Therefore:

$$\text{LogMAR} = -\log 0.32 = 0.5$$

Also, 6/60 in Snellen's chart means that decimal acuity = 0.1, and:

$$\text{logMAR} = -\log 0.1 = 1$$

The values of logMAR are the same when Snellen's chart values are in

meter or foot because it gives the same decimal acuity. (Table 8.3b).

Table 8.3: LogMAR scores. a- logMAR visual acuity scales. b- logMAR versus Snellen chart values.

a- logMAR visual acuity scales b- logMAR versus Snellen chart values.

-a-

-b-

Visual acuity scales

Foot	Metre	Decimal	LogMAR
20/200	6/60	0.10	1.00
20/160	6/48	0.125	0.90
20/125	6/38	0.16	0.80
20/100	6/30	0.20	0.70
20/80	6/24	0.25	0.60
20/63	6/19	0.32	0.50
20/50	6/15	0.40	0.40
20/40	6/12	0.50	0.30
20/32	6/9.5	0.63	0.20
20/25	6/7.5	0.80	0.10
20/20	6/6	1.00	0.00
20/16	6/4.8	1.25	-0.10
20/12.5	6/3.8	1.60	-0.20
20/10	6/3	2.00	-0.30

logMAR	SNELLEN (FOOT)	DECIMAL	SNELLEN (METRIC)
1.5	20/640	0.03	6/192
1.4	20/500	0.04	6/152
1.3	20/400	0.05	6/120
1.2	20/320	0.063	6/96
1.1	20/250	0.08	6/76
1.0	20/200	0.10	6/60
0.9	20/160	0.125	6/48
0.8	20/125	0.16	6/38
0.7	20/100	0.20	6/30
0.6	20/80	0.25	6/24
0.5	20/63	0.32	6/20
0.4	20/50	0.40	6/15
0.3	20/40	0.50	6/12
0.2	20/32	0.63	6/10
0.1	20/25	0.80	6/7.5
0.0	20/20	1.00	6/6
-0.1	20/16	1.25	6/5
-0.2	20/12.5	1.60	6/3.75

2- Slit lamp biomicroscope:

It is a device consists of a high intensity source of light with a thin slit in conjunction with a biomicroscope. It is used to examine the anterior and posterior segments of the eye which include, in order, the eye lids, lashes, tear film, conjunctiva, cornea, aqueous humor, iris and pupil, lens, and vitreous humor, where another hand held, lens is used to examine the retina. (Fig 8.16).

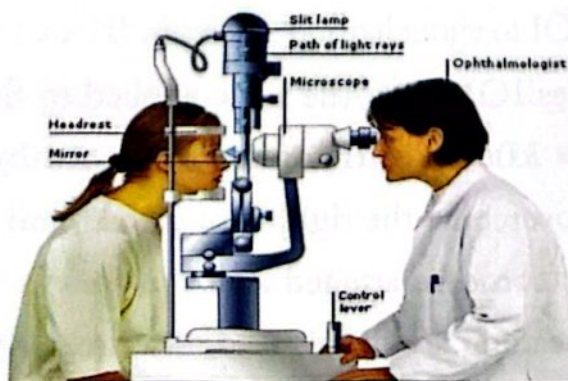


Fig 8.16: Slit lamp biomicroscope.

The slit can be opened more to get more illumination of the eye as required by the examiner.

There are no special preparations required for adults in the slit lamp test, but the children probably require some preparations depends on their medical conditions.

During the exam, the patient will be asked to place his chin in the chin rest and his forehead against the forehead band in the machine, to keep his head steady during the exam. The examiner may use eye drops, contain a yellow dye, to help the examiner to clarify the problem in the front of the eye. Also, dilating drops (which lasts 15-20 min to work) are used to widen the pupil to see the back of the eye more clearly. Then the examiner will look up the parts of the eye to detect abnormalities and diseases in the eyes.

The slit lamp can detect many abnormalities and diseases in the eyes, such as the keratoconus, cataract conjunctivitis (pink eye), Fuchs dystrophy (fluid buildup in cornea causing it to swell and thicken), cornea swollen or ulcer, diabetic retinopathy, macular degeneration, retinal detachment, retinal vessel occlusion and retinitis pigments.

3- Tonometer:

It is a device used to measure the fluid pressure in the eye i.e., the intraocular pressure (IOP), which is normally measured in mmHg.

The glaucoma can lead to blindness especially for people of age > 60 yrs, thus, the tonometry is very important because it can determine whether the patient is at risk of glaucoma.

Basically, the tonometer measures IOP from the force applied to flatten an area of cornea of a diameter = 3.06 mm. This force is resisted by the surface tension of the tear film to overcome the rigidity of cornea, and $IOP = \text{amount of the applied force per corneal flattened area}$.

It uses a slit lamp for accurate reading and a prism to measure the force needed to flatten the cornea.

There are several types of tonometer such as the contact, air puff (non-contact), and automated. (Fig 8.17).

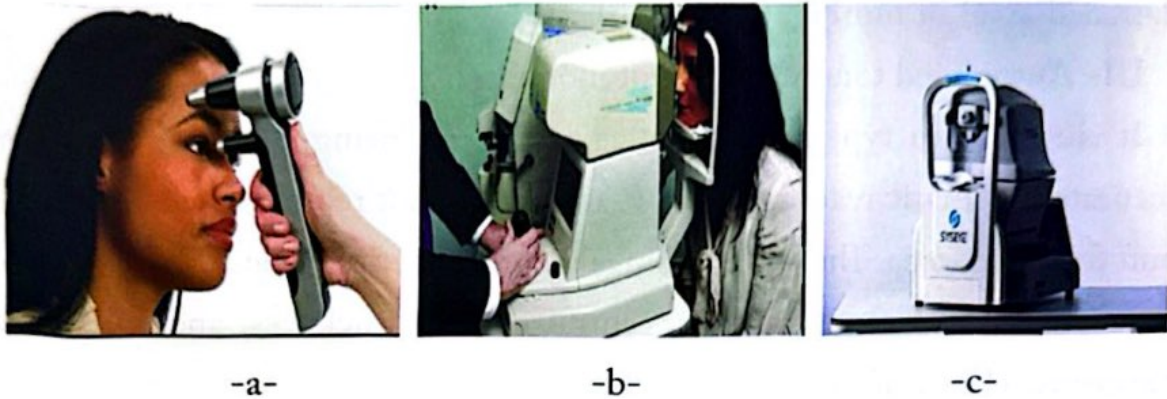


Fig 8.17: Tonometer types. a- contact tonometer. b- noncontact tonometer. c- automated tonometer.

I- Contact tonometer:

It is a type of tonometer in which the tip of tonometer touches the cornea of the patient. The patient requires anesthetic eye drops because this tonometer is directly in contact with his cornea.

The contact tonometer is based on the Goldman application tonometry (GAT), in which the tonometer touches the cornea to find the amount of force required to flatten the cornea.

In this process, the tension knob will be turned on, and the corneal force is altered until the internal aspect of two semicircles contact each other, which indicate the IOP.

A handheld version of the contact tonometer, without a light slit, is also used to measure IOP, where the normal range of IOP = 10-21 mmHg. (Fig 8.17a).

The modern contact tonometer is digital which facilitates easy, fast, and accurate measurement of IOP, even in the very dark examination room.

II- Noncontact (air puff) tonometer:

This tonometer does not contact the cornea, and the patient does not require anesthetic drops. It uses a column of air of gradually increasing intensity (air buff), which is shut off at the point of corneal flattening.

The force required to flatten the cornea is detected by sensors to determine the IOP, and this force is recorded and converted to pressure (force/corneal flattened area) in mmHg. (Fig 8.17b).

III- Automated tonometer (autotonometer):

It is a modern type of noncontact tonometer using a computer to give accurate automatic readings of IOP, and it does not require anesthesia or air puff for the cornea. The IOP readings are dependent on the surface tension, tissue tension, corneal stiffness, central corneal thickness, and astigmatic curvature. (Fig 8.17c).

4- Retinoscope (skiascope):

Retinoscope is a handheld instrument used for the test of eye refractive defects, which projects a beam of light vertically and horizontally across the eye of the patient, mostly an infant, child handicapped, or with severe intellectual communication problem. It is used to determine the eye defects (myopia, hypermetropia, and astigmatism), and the required glasses to correct the defects.

A streak, or a spotlight, is produced from a filament at the base of retinoscope handle, and it is controlled by the retinoscopist, who monitors the reflex through a slit in the mirror.

The eye test by retinoscope is called retinoscopy or skiascopy. The examiner observes the movement of reflex of a reddish orange light, which is emitted from the retinoscope, and reflected from the back of the eye.

During retinoscopy, there are four states of light focusing which indicate the eye defect, which are:

I- On retina, this means that the patient has no refractive error.

II- In front of retina, it means that the patient has myopia.

III- Behind retina, this means that the patient has hypermetropia.

IV- On retina but in more than one focus, the eye has astigmatism.

Therefore, when the eye has a refractive error, different trials of lenses powers are introduced in front of the eye, using a trial frame, until reaching a lens

power that neutralizes the reflex (reflection from one focus on retina) which indicates the required lens power for the eye defect of the patient. (Fig 8.18).



Fig 8.18: Retinoscope device.

The eye pupil of the patient is dilated by certain eye drops for widening the pupil to determine the eye defect more accurately. The eye drops are used because it is easier to see the reflex when the pupil is large, and the drops temporarily limit the ability of the eye to focus or accommodate.

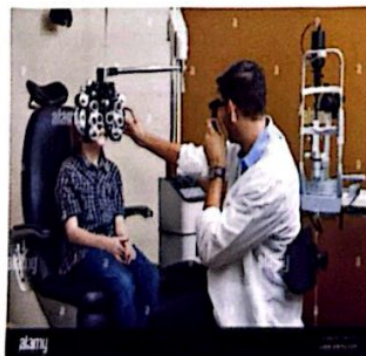
5- Phoropter (refractor):

It is a special machine, contains a variety of multiple lenses, to measure the refractive error, and test the binocular vision (how well the two eyes work together). (Fig. 8.19a).

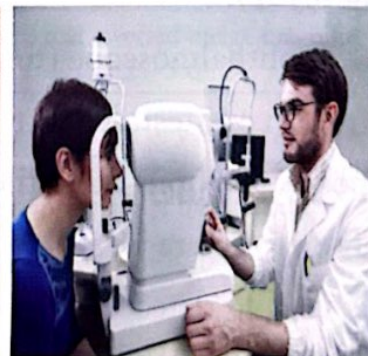
If a patient had an eyesight test indicated a refractive problem like near sightedness, far sightedness, or astigmatism, then the next step, he will take phoropter test to get an exact vision correction.



-a-



-b-



-c-

Fig 8.19: The phoropter. a- phoropter device. b- phoropter and retinoscope test for patients who can't give signs. c- autorefractor.

The procedure of phoropter test can be summarized as follows:

I- The patient sits behind the phoropter and looks through its lenses to an eye chart, normally Snellen's chart, or another chart.

II- The distance between the patient and the chart = 6 m (20 ft) for eye refractive error tests, and 40 cm (16 inch) for reading glasses test.

III- Lenses are used to refract the light on retina of the patient eye, where these lenses have optical power increment measured in 0.25 D.

IV- The examiner changes the lenses of phoropter, and ask the patient, until the patient gets his best vision. This represents his sight prescription to determine the spherical power for near sightedness, far sightedness, old sight, reading glasses, or cylindrical power and its axes for astigmatism.

V- The phoropter is normally used with the retinoscope for sight test of the people who cannot give signs to the examiner when they are asked about their vision at the time when phoropter lenses are changed. (Fig 8.19b).

VI- Another type of refractors, a computerized refractor called automated refractor or autorefractor, is used to measure the refractive errors, prescription for glasses or contact lenses, and the change of light as it enters the eye. The measurement is performed in the meridians of the eye, the cylinder, sphere, and eye refraction axes, which are calculated by the autorefractor. (Fig 8.19c).

6- Ophthalmoscope (fundoscopy):

It is an eye test device to see the inside of eye fundus (the part of eyeball opposite to pupil) and other eye structures.

The procedure of eye test using ophthalmoscope is called ophthalmoscopy. It is used to detect retinal vascular disease, glaucoma, swollen optic disc that increases with increase of intracranial pressure (which could be due to benign intracranial, hydrocephalus or brain tumor). Also, it is used for regular eye test, once every 6-12 months, for patients with diabetic retinopathy. (Sec 9.20).

For better inspection by eye ophthalmoscopy, tropicamide eye drops are used to dilate the pupil of the eye.

There are two kinds of ophthalmoscopy, the direct ophthalmoscopy, and indirect ophthalmoscopy which is more popular. (Fig 8.20).

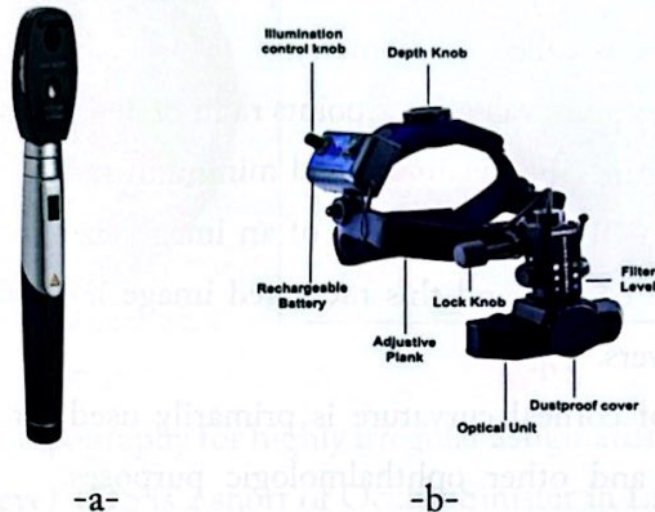


Fig 8.20: Ophthalmoscopy. a- direct. b- indirect.

Each of the direct and indirect ophthalmoscopy has certain characteristics, and it is used in different applications. (Table 8.4).

Table 8.4: Comparison between the direct and indirect ophthalmoscopy.

	Direct ophthalmoscopy	Indirect ophthalmoscopy
1	Simple technique, does not require a dilated pupil, and it is held by the examiner hand directly at the level of the patient eye.	More complicated technique, requires a dilated pupil, and worn on the head of the examiner.
2	Produces virtual erected image magnified about 15 times.	Produces real inverted image magnified about 2 – 5 times.
3	It has a narrow field of view because the examination distance is as close as possible to the eye of the patient.	It has a wide field of view because the examination distance is as an arm length from the eye of the patient.
4	Illumination is not bright, not useful in the hazy media, and does not permit to examine moderate cataracts or to see more retinal details.	Illumination is bright, useful in the hazy media, and permits to examine moderate cataracts or to see more retinal details.
5	It does not permit binocular vision (not stereoscopic vision).	It permits binocular vision with depth perception (stereoscopic vision).

6	It is useful for the routine physical examinations.	It is useful for peripheral viewing of retina.
---	---	--

7- Keratometer (ophthalmometer):

It is a device used to measure the front curvature surface of cornea of a radius $\approx 3-4$ mm i.e., it measures the power of outer surface of the cornea. This procedure is called keratometry.

Keratometry gives values of 2 points radii of 3-4 mm zone at the anterior corneal curvature (the maximum and minimum radii) with an angle of 90° between them. The measurements of an image size are a result of these 2 points on the cornea, and this measured image is different for corneas of different powers.

The data of corneal curvature is primarily used for detecting irregular astigmatism, and other ophthalmologic purposes.

The keratometer has two types, the manual keratometer and autokeratometer. They can measure the curvature of cornea in mm or the power of cornea in diopter by measuring the vertical meridian, horizontal meridian, and the axes, (Fig 8.21).



-a-



-b-

Fig 8.21: Keratometry device. a- manual keratometer. b-autokeratometer. The manual keratometry is preferable in some exams for specific reasons. Keratometer was developed by the German physician and physicist, Herman Von Helmholtz, (1821-1894).

8- Corneal topography (photokeratoscopy or videokeratographer):

It is a computerized imaging technique used for 3D mapping of thousands

points of outer curvature for the entire of cornea, and displays the exact shape of cornea in few seconds with no pain. (Fig 8.22).

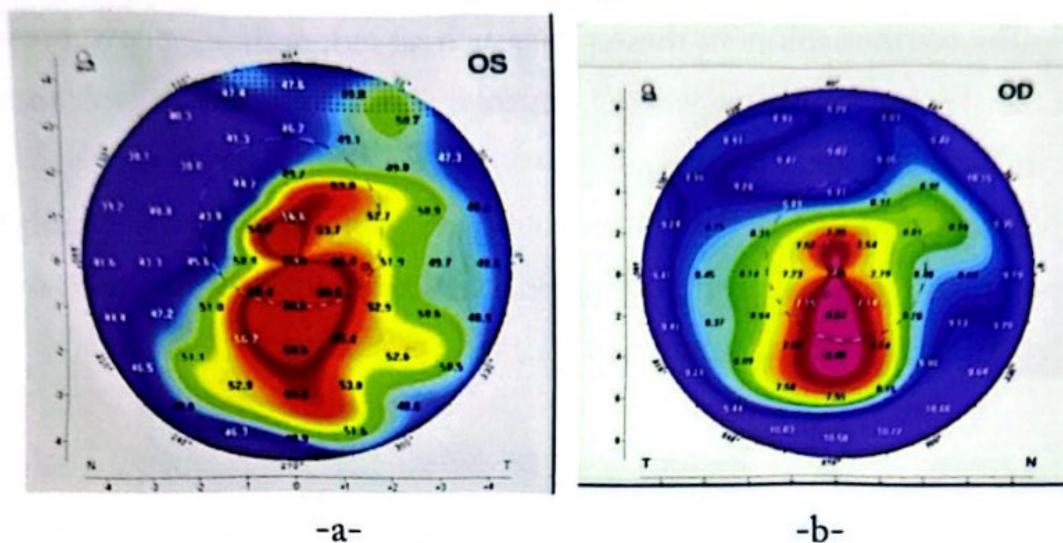


Fig 8.22: Corneal topography for highly irregular astigmatism. a- OS (left eye). b- OD (right eye). (OS is a short of Oculus Sinister in Latin language which means left eye, and OD is a short of Oculus Dexter in Latin language which means right eye).

The 3D map, used since 2013, is a valuable aid in the diagnosis and treatment of a number of sight conditions, such as the planning of laser refractor surgery (lasik, lasek, or PRK), cataract surgery, and intraocular lens (IOL) implantation. Also, it is used in assessing of several cases such as the corneal transplant management of corneal scarring, evaluation of astigmatism, and dry eye.

The procedure of corneal topography is performed as follows:

I- The patient is seated facing the device, where the device can be raised to the level of patient's eyes.

II- The light is focused from the device on the front surface of the cornea, which is reflected at several thousand points across the cornea to a digital camera in the device.

III- The computer of the device analyzes this data to determine the positions and heights of these points and plot the topographical map of

cornea curvature which is directly measured in diopter. These plots are performed in several graphical formats like a sagittal map.

IV- The warmer colors in these plots, such as red, orange, yellow, brown, and tans (thinking of longer wavelength colors or existing fire and volcanoes) indicate steeper curvature.

V- The cooler colors in these plots, such as the green, blue, violet (thinking of shorter wavelength colors or calming blue water) indicate flatter curvature in the cornea. (Fig 8.23).

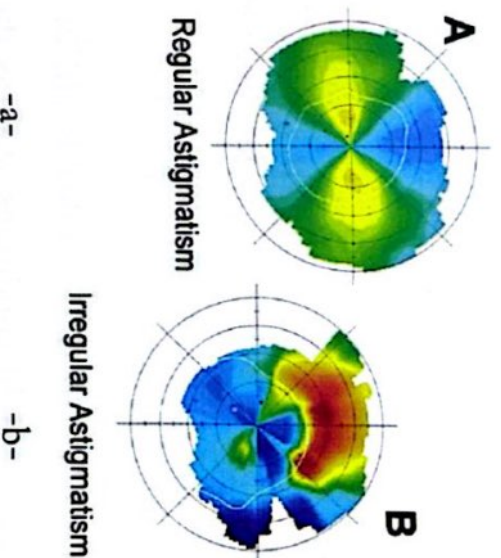


Fig 8.23: Corneal topography of astigmatism with warmer and cooler colors. a- regular astigmatism. b- irregular astigmatism.

8.13- Endoscopes:

The endoscope is an optical device of a long thin flexible tube (slender tubule) or optical fiber, with a light and camera put on the head of the tube, and mostly equipped with a television screen, to show the image of inside of the hollow organs or cavities of the body. It is inserted into the body through an opening such as the mouth, rectum, urethra, and vagina, or small incision in the body. (Fig 8.24).

Some kinds of endoscopes are equipped with surgical tools, like forceps or scissors, to take a sample from the tissues called biopsy, and some other kinds used in surgery called laparoscopes.

The laparoscope is a fiber optic endoscope, inserted through small incisions

in the abdominal wall to view the organs in the abdomen, and permit a small-scale surgery.

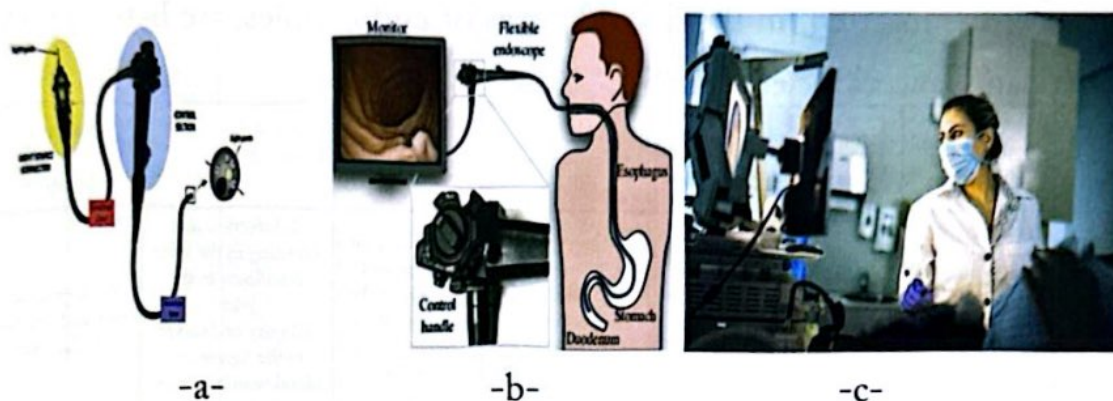


Fig 8.24: Endoscopy. a- an endoscope device. b- conventional gastroscopy. c- patient endoscopy.

The quality of endoscope visualization depends on the resolution (ability to distinguish between two adjacent points or objects i.e., the ability to discriminate details) and magnification of the image.

The procedure of using the endoscope is called endoscopy and that of using laparoscope is called laparoscopy.

Typically, there are 3 major uses of endoscopy in the hollow organs or cavities of the body which are:

- 1- Diagnosis, to determine the reason of any abnormal symptoms.
- 2- Biopsy, to remove a small sample of suspicious using forceps and scissors or a needle biopsy, when an initial test suggests abnormal tissue (lesion), mass, or tumor which can be then sent to the lab for investigation.
- 3- Surgery, for treatment like using of laparoscopes for a stomach ulcer, removing of a tumor mass, or removing of gall stones

There are several types of endoscopies which are categorized relative to the investigated area of the body. (Table 8.5).

Table 8.5: Comparison between the types of endoscopies.
 [In this table, the first three columns (type, area examined, and where the endoscope is inserted) in the first 13 types of endoscopies, are listed by the American Cancer Society (ACS)].

No	Type	Area examined	Where scope is inserted	Additional preparation for the patient	Period of procedure	Risks	Reasons for having endoscopy
1	Arthroscope	Joints	Through a small incision near the joint.	1-Stop taking aspirin or ibuprofen 6-12 weeks before. 2- stop eating and drinking 6-12 hrs before.	Depending on: 1-type and severity of the joint problem. 2-complexity of the procedure.	1-Infection and bleeding in the joint 2-stiffness in the joint 3-injury or damage to the ligaments, blood vessels, nerves, and cartilages.	Pain, swelling, cyst and fracture in the joint
2	Bronchoscopy	Lungs	Into nose or mouth	1-Stop eating and drinking 6-12 hrs before. 2- using 1 or 2 enemas about 2 hrs before	30 min	1-Trouble breathing. 2-bleeding 3-infection.	Lung disease, tumor, infection, and chronic cough.
3	Colonoscopy	Colon	Through the anus	1-Drink a clear liquid diet at 24-72 hrs before procedure. 2-stop eating and drinking 6 hrs before procedure 3-inflation of the colon by CO ₂ during the procedure.	30 – 60 min	bleeding and tear in the rectum or colon walls.	1-The reason of unexplained changes in the bowel habits, chronic constipation, and diarrhea. 2-symptoms of pain or bleeding in the abdomen. 3-signs of tumor and cancer in the colon.
4	Cystoscopy	Bladder	Through the urethra	1- Urine test 2- taking antibiotics 3-stop some medications	15– 30 min	1-Infection 2-swollen urethra (urethritis) 3-bleeding	1-Overactive bladder. 2-urinary tract infections. 3-blood in urine. 4-pelvic pain. 5-diagnosis of enlarged prostate, blockage, tumors, and cancers in the urinary system. 6-remove small bladder tumors and stones. 7- biopsy.

5	Enteroscopy (With one or two balloons)	Small intestine.	Through the mouth or rectum	1-Stop drinking and eating 8 hrs before 2-stop taking blood thinning, iron supplement and peptobismol.	45/120 min	1-Nausea 2-severe throat 3-abdominal bloating 4-mild cramping	1-Blockage, damage, bleeding, and tumors in the intestine. 2-severe diarrhea, unexpected malnutrition, abdominal X-rays result, and high white blood cell count.
6	Hysteroscopy	Inside of the uterus	Through the vagina	1-Clean vagina with antiseptic 2-stop food and drink 8 hrs before, for general anesthesia. 3-pregnant woman is not allowed for this test.	5 - 10 min	1-Scaring of the uterus. 2-infection. 3-bleeding.	1-Malformation of uterus 2-prior to dilation and curettage procedure. aid in the removal of fibroids and polyps 3-conjunction with a laparoscopy
7	Laparoscopy	Abdominal or pelvic area	Through a small incision near the examined area	1-Stop food and drink 6-12 hrs before. 2-stop taking blood thinning medication few days before.	30-60 min	1-Infection 2-bleeding 3-damage to the abdominal organs	1-Diagnosis of abdominal organs where the imaging technique (sonar, CT scanner and MRI) are unable to do that. 2-taking biopsy from these organs. 3-detection of liver disease, fluid in the abdominal cavities, abdominal masses or tumor and cancer progress in the abdomen

8	Laryngoscopy	Larynx	Through the mouth or nostril	Stop food and drink for 8 hrs before, for cases requiring more than mild anesthesia	5 - 45 min	Minor irritation in the throat	1-Persistent or bloody cough. 2-persistent carache. 3-difficult swallow, throat pain or bed breath. 3-mass or growth in the throat. 4-hoarseness (abnormal voice change)
9	Mediastinoscopy	Medastinum (area between the lungs)	Through an incision above the breastbone	Blood test for complete blood count or clotting factors	60 - 90 min	1-Allergic reaction to the anesthesia 2-hemorrhage (which can be vital due to accidental puncture of blood vessels).	1-Taking biopsies from the lymph nodes and the area between lungs for cancer test. 2-identify infection like tuberculosis. 3-checking spread of lung cancer and lymphatic cancer including (Hodgkin's lymph disease).
10	Sigmoidoscopy	Rectum and lower part of the large intestine known as the sigmoid colon	Into the rectum	1-Using one or two enemas about 2 hrs before. 2-taking clear liquid diet for 1-3 days before, it may mix with powder laxative to help emptying the intestines.	16-20 min	Treating the wall of colon or rectum which cause bleeding	1-Checking for abnormal cells, polyp's ulcer, and cancer biopsy. 2-abnormal pain, rectal bleeding, changes in the bowel habits and unexpected weight loss.
11	Thoracoscopy, also known as pleuroscopy	Area between the lungs and the chest wall.	Through a small incision in the chest	1-Stop eating and drinking for the time that the doctor tells, 2-taking a bath before come in.	2-3 hrs	Very rarely pain, infection and bleeding	1-Trouble breathing. 2-coughing up blood. 3-removing excess fluid around the lungs. 4-biopsy. 5-treat small lung cancers 6-surgery to treat lung cancers, 7-lung volume reduction. 8-esophageal disorders. 9-surgery to remove part or all the esophagus, or thymus gland

12	Upper gastrointestinal endoscopy, also known as an esophagogastroduodenoscopy	Esophagus and upper intestinal tract	Through the mouth	1- Stop eating and drinking 4-8 hrs before. 2- stop taking blood thinning few days before according to the doctor instructions.	5-10 min	Small risk of: 1-cause a small hole in the stomach, esophagus, or small intestine 2- prolonged bleeding in biopsy.	1- Pain in upper abdomen or at swallowing. 2- vomiting blood. 3- feeling of fullness, or food is lodged behind breastbone after eating. 4- severe chronic heart burn. 5- unexpected anemia or weight loss. 6- treatment of pelvic ulcers, cirrhosis (liver long term disease) Crohn's disease (inflammatory bowel disease), swollen veins in the lower esophagus. 7- biopsy
13	Uretroscopy	Ureter	Through the ureter	Need to pee before	30 - 45 min	Small chance of bleeding, injury, or infection in ureter	Diagnose of removing tumor, abnormal tissue, polyps, or stones in the urinary tract. 2- urine blockage.
14	Capsule endoscopy (small pill with tiny camera)	Intestine using recording device on a belt worn around the waist of patient	Swallow by the mouth	Taking laxatives before the endoscopy	8 hrs	Capsule stuck in a narrow passage causes bleeding, fever, and vomiting	1- Abdomen pain. 2- unexpected bleeding. 3- gastrointestinal ulcer and cancer
15	Chromoendoscopy (using stains or dye mostly methylene blue)	1- Esophagus 2- colon	Mouth	1- Taking a clear liquid diet several days before endoscopy. 2- taking laxatives or enemas before endoscopy	11-20 min	1- Impractical use of the dye causes permanent stains in the contacted surface 2- changes the color of urine into blue color.	1- Monitoring of the esophagus for Barrett's esophagus and dysplasia in the inflammatory disease. 2- increase of the detection rate for various pathological processes. 3- high differences in mucosa, dysplasia, and malignant changes.

16	Narrow band imaging using special e-filter to give blue and green light	Digestive track (mucosal and blood vessels)	Mouth	1-Dietary restriction. 2-polyethylene glycol regime. 3-flush pump to clear view mucosa and bleeding source	5-20 min for each sub exposure	1-Creation of more contrast between blood vessels and mucosa. 2-accurate assessment of lesions. 3-differentiating between premalignant and malignant lesions. 4-detection of malignant transformation 5-Detection of early neoplasia	Improve detection of lesions in the digestive track
17	Endoscopic mucosal resection (needle through endoscope)	Liquid injection underneath abnormal tissue in the digestive system	Mouth or rectum	1-Stop taking the diabetes medication, blood clotting or that interfere with sedatives. 2-fasting and no smoking 8 hrs, before. 3-cleaning the colon by enema or liquid laxative (if the procedure involves colon).	1-2 hrs	1-Bleeding, perforation (a hole in the GIT). 2-narrowing the esophagus. 3-puncture through the wall of GIT.	1-To easily remove the precancerous, early-stage cancer, and other abnormal tissues, from the digestive track. 2-accurate local stages of the disease. 3-treatment of Barret's dysplasia in the esophageal squamous dysplasia, and the colorectal adenoma
18	Endoscopic ultrasound	1-GIT 2-lungs 3-liver 4-pancreas 5-lymph nodes	Mouth or rectum	1-Taking laxative before the procedure. 2-follow a liquid diet	30-60 min	Infection 2-bleeding. 3-pancreatitis 4-tearing of the intestinal wall or throat	Evaluation of: 1-Barret's esophagus lymphoma. 2-bile duct stones. 3-sarcoidosis (disease affect lungs and lymph glands together). 4-pancreatic cyst. 5-neuroendocrinal tumors. 6-GI cancer, esophagus, stomach, colon, and pancreas. 7-rectal ampullary (joint of bile duct). 8-lungs cancer.